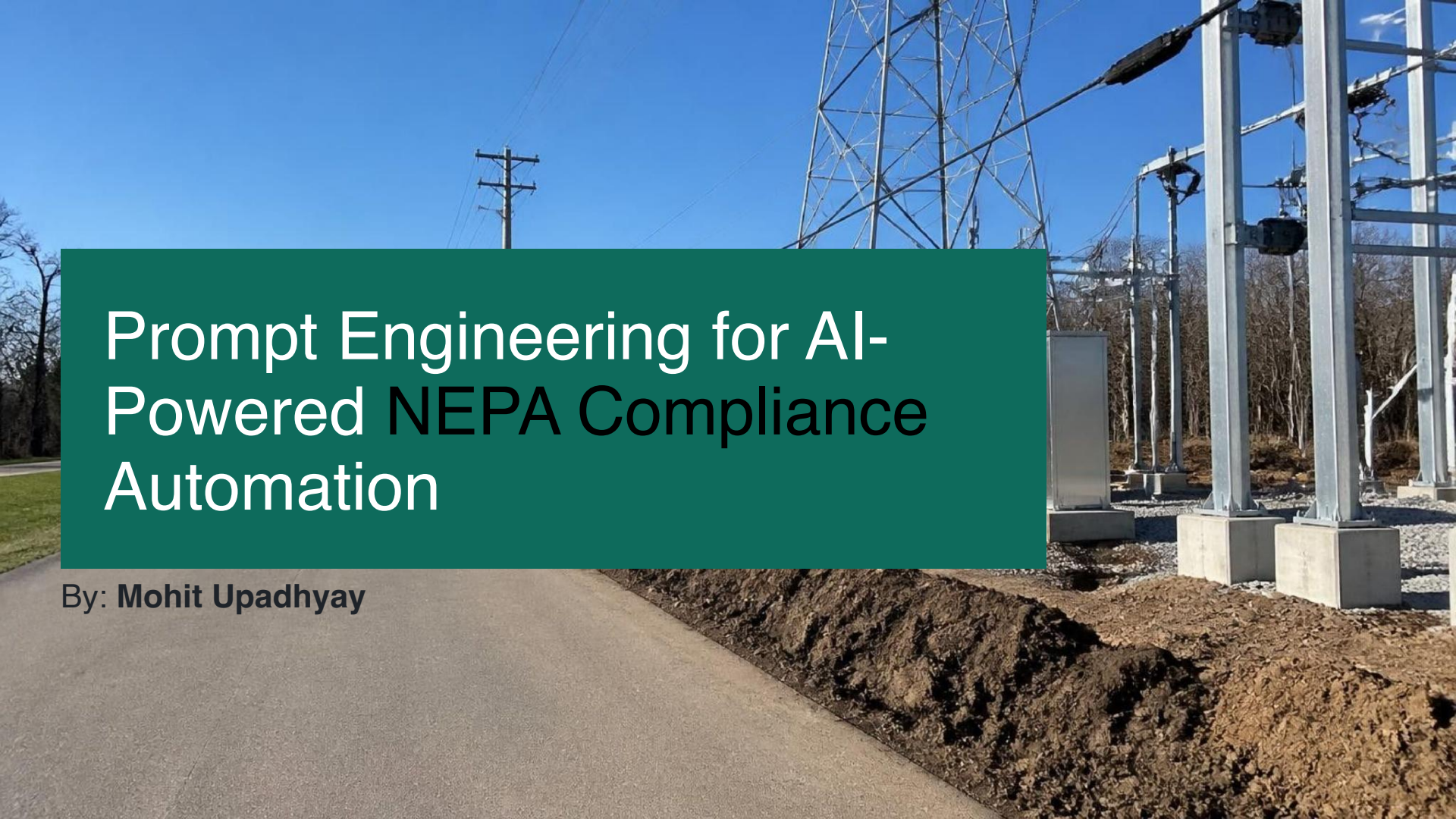


Prompt Engineering for AI-Powered NEPA Compliance Automation

By: **Mohit Upadhyay**



Permitting is an evidence problem, not a document problem

- Routine-looking installations touch cultural resources, ground disturbance, visual character, and agency authority.
- Link the action to the right legal framework and a defensible record.

DESIGN PRINCIPLE

Evidence-first automation

- Build an evidence-navigation and drafting system, not an autonomous regulator.
- Software accelerates collection and explanation; reviewers own interpretation and approval.
- A fluent answer is never proof a categorical exclusion applies.

A categorical exclusion starts the inquiry, it doesn't end it

- Identify the federal action, responsible agency, current procedures, and facts that could defeat the exclusion.
- Section 106 is a separate process. Keep separate status fields for NEPA and historic preservation.
- Never call a NEPA pathway complete while other reviews are unresolved.

Separate status fields for NEPA and historic preservation.

Not all legal sources carry equal authority

Statutes, regulations, agency procedures, guidance, consultation records, and site determinations rank differently.

- Supply jurisdiction, agency context, and effective dates in prompts.
- Clarify federal nexus and historic-property status first.

One evidence model that keeps provenance

PROJECT FACTS

Footprint, excavation, elevations, access, utilities

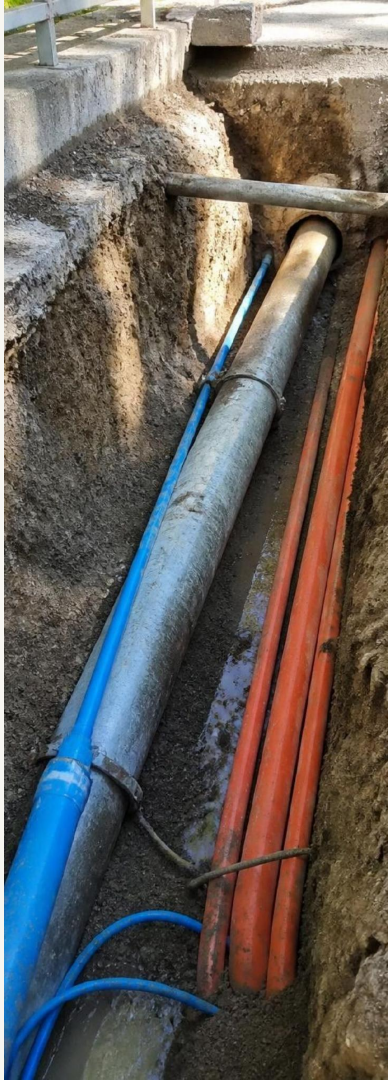
REFERENCE MATERIAL

Retrieved material kept separate from project facts

EACH FACT KEEPS PROVENANCE

Source file · page or geometry · date · extraction method · reviewer status

Geospatial records retain coordinate system, precision, and spatial-operation history.



Integrations come with source-specific limits

Street View, NPS/IRMA, National Register, FCC ULS, EPA ECHO, USGS 3DEP, and SHPO systems each cover only part of the picture.

Document owner, cadence, coverage, omissions, usage limits, and fallback for every source.

Never turn missing data into a false clearance

Log retrieval time, query parameters, identifiers, response versions, and failures.

True negative

Distinguish from a failed query or inaccessible record.

Failed query / inaccessible record

Authoritative source unreachable → request manual evidence.

Prompts as controlled analytical instructions

- Limit the system role to identifying rules, extracting facts, surfacing uncertainty, and drafting for review.
- Forbid invented citations, unsupported site findings, and claims of granted approval.
- Separate observed facts, inferred facts, open questions, and applicable rules.
- A second verification prompt checks that citations support the conclusion.

A structured output schema plus abstention rules

Question	Controlling authority
Observed facts	Citations
Missing evidence	Reviewer action
Approval status	

- Escalate on conflicting drawings, uncertain dates, or disputed historic effects
- Version-control prompts and test them on curated cases

Retrieval grounds the model in current law

1. Filter by jurisdiction, agency, document status, and effective date before searching.
2. Hybrid semantic and keyword search catches definitions, exceptions, and cross-references.
3. Chunk clauses with headings, definitions, and neighbors so exceptions travel with rules.
4. Citation checker verifies each legal claim; conflicts are flagged, never averaged.

Vision and spatial analysis flag conflicts

- Read title blocks, legends, scale, trench paths, and elevation annotations; engineers verify material measurements.
- Compare disturbance polygons with parcels, resources, and elevation models.
- A buffer or overlap is a screening signal, not an effects determination.
- Label every machine-derived feature with confidence or review status.

A human-supervised workflow with auditable steps

01

Intake validation

Classification

02

Rule retrieval

Evidence requests

03

Site conditions dossier

- Rules engine handles measurable conditions; the language model explains and drafts.
- Route issues to environmental, historic-preservation, and engineering reviewers.
- Designated approval required before any external submission.

Audit, security, and how to roll out

- Reconstruct any recommendation from revision, prompt version, model config, passages, tool responses, and approvals
- Role-based access; treat retrieved web content as untrusted and test for prompt injection
- Start with one narrow project category; judge by source accuracy, omissions, escalation quality, and correction burden
- Faster retrieval stays a hypothesis until measured against a baseline



Success is a more reviewable decision record

- Help experts find evidence and see what remains uncertain, not declare projects compliant.
- Principles transfer to other high-stakes workflows, but each regime needs fresh legal mapping and evaluation.

Thank you

Questions and discussion are welcome.